ITERYX ‘26

Problem Statement Number	-	NO.2
Problem Statement	-	AUTOMATED GRIEVANCE CATEGORIZATION & ROUTING SYSTEM
Domain	-	AI & Machine Learning
Team Name	-	HORCURX
Team LeaderName	-	DEJASWINI B G
College Name	-	RAJALAKSHMI ENGINEERING COLLEGE

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AI-Based Automated Grievance Categorization & Intelligent Routing System

USER COMPLAINTS

AI CLASSIFIES

HIGH | MID | LOW → ESCALATION

CORE SYSTEM

- **Auto classify & route complaints**
- **Detect priority (High / Medium/ Low)**
- **Track complaint status**
- **Multilingual input support**

BENEFITS

- Removes **fake & duplicate** complaints
- Prioritizes **genuine issues**
- **Reduces** response delay
- Improves efficiency

NOVALITY

1.Spam Detection

- Detects duplicate / irrelevant complaints
- Prevents misuse

2.Smart Escalation

- Auto escalates
- unresolved issues
- Tracks delays

TECHNICAL APPROACH

Client-Side Technologies

React JS

Server-Side Technologies

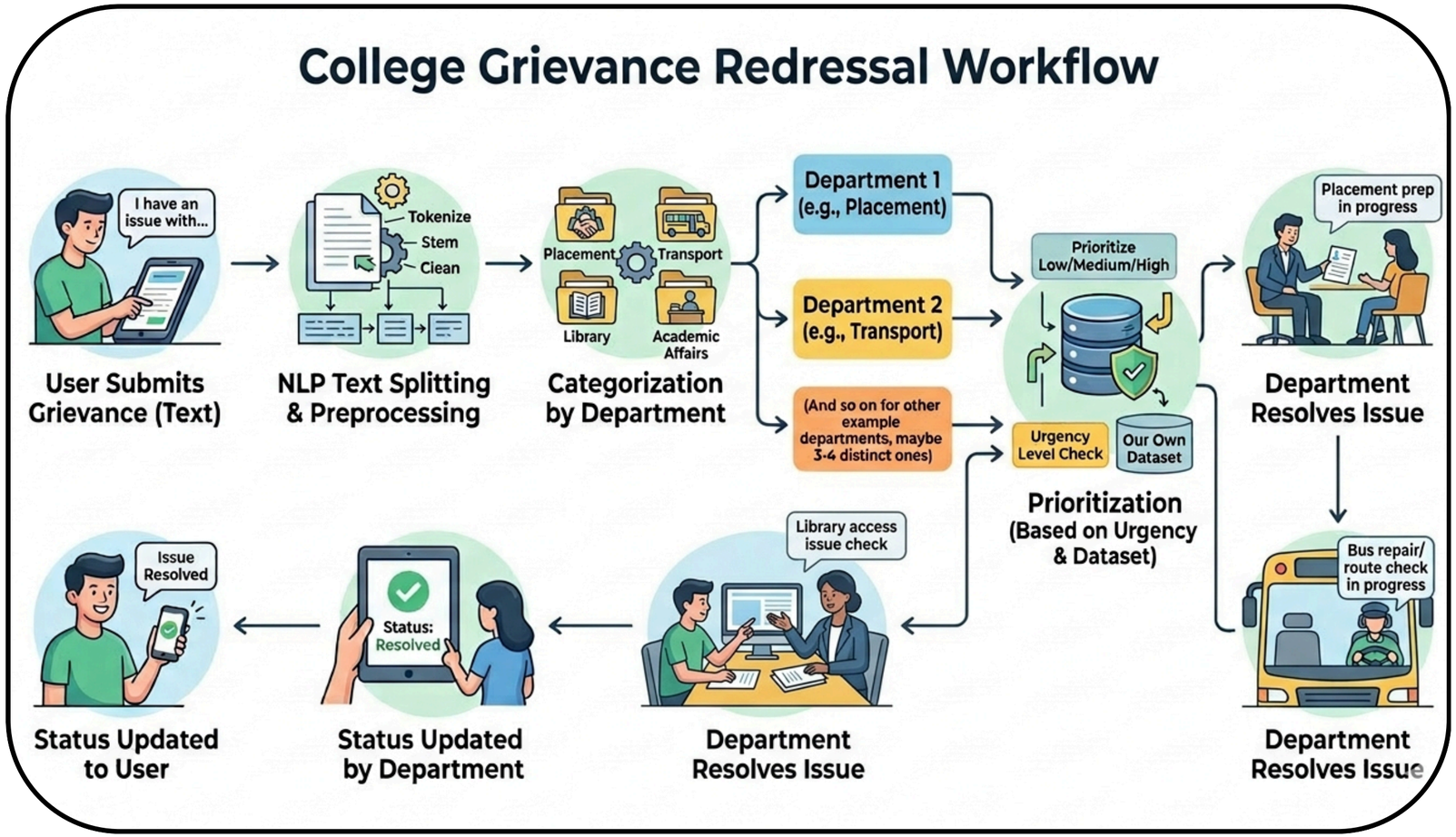
node JS

API handling

Express JS

Database

mongoDB



FEASIBILITY AND VIABILITY

Despite challenges, the system remains practical and scalable with proper AI optimization and safeguards

Feasibility Analysis

Technically Feasible
→ Uses proven NLP models like BERT for accurate classification

Economically Feasible
→ Low-cost using open-source tools and cloud services

Potential Chalenges & Risks

Multilingual Accuracy Issues
→ Translation errors may affect classification

Data Privacy & Security Risks
→ Sensitive user information must be protected

Strategies to Overcome Challenges

Data Preprocessing & Cleaning
→ Removes noise and improves input quality

Secure Data Handling
→ Encryption & authentication mechanisms

IMPACT AND BENEFITS

IMPACT

Students (Users):
- Fair, fast and accessible complaint resolution

Administration:
- Reduced manual workload & improved efficiency

Departments:
- Accurate and organized complaint handling

Management:
- Better decision making using data insights

BENEFITS

Social Benefits:
- Improves transparency and accountability
- Ensures safe reporting (especially harassment cases)

Economic Benefits:
- Reduces operational cost and manpower effort
- Saves time through automation

System Benefits:
- Spam detection & auto escalation
- Efficient resource utilization

LIMITATION

- Depends on dataset quality
- Possible misclassification of complex inputs
- Requires initial data preparation

RESEARCH AND REFERENCES

PAPER RESEARCH:

1. https://ijrpr.com/uploads/V6ISSUE11/IJRPR55290.pdf
2. https://ijirt.org/publishedpaper/IJIRT169187_PAPER.pdf

Existing Automated Grievance Solutions

1. **CPGRAMS (Centralized Public Grievance Redress and Monitoring System)**
2. **Smart City Grievance Management Systems**
3. **CRM Complaint Management Systems(Especially for Salesforce and ZOHO helpdesk)**